

CMS Submission Infrastructure: strategic goals and challenges

- Scalability and stability
 - Continuous evolution of the infrastructure to keep up with increased CMS resource needs, while keeping it stable and under efficient utilization
- Adoption of new technologies
 - Token-based authz: almost completed challenge
 - Secrets management improvements
 - IPv6: transition to dual stack almost completed in the SI
 - Containerized services: use of containers could ease deployment of horizontally scalable SI components
 - Consolidation of our services configuration repositories
 - Evaluate new technologies for allocating and using resources (e.g. Decision Engine)
- Resource Heterogeneity
 - Efficient use of GPUs
 - Advancing on the required description granularity for resources and requests
 - Multi-step jobs, e.g. via Inference servers launched by the pilots to serve multiple jobs on the same node (?)
 - Keep supporting and improve multiple architectures provisioning (arm, powerpc, ...)
- Continue the integration of HPC centers into SI
 - Impact on increasing scale and heterogeneity of resources under SI
 - Challenging operational model: outside of usual (WLCG) framework of pledges and standard slot requirements (VOCARD)
- Monitoring and debugging goals
 - Enhanced monitoring of SI pools performance and scalability limits
 - CPU usage monitoring
 - Payload scheduling inefficiencies
 - Internal payload CPU use profiling by monitoring pilots
 - Re-Enabling condor_ssh_to_job debugging tool with containerized slots
 - Goal: condor_ssh_to_job to GPU slots!
 - Deploy historical monitoring for SI (new producer to monitor infrastructure)